

Seborrhoeic Dermatitis and Skin Microbiome

Loci GWAS of seborrheic dermatitis, innate immunity and fungi (*Malassezia*, *Candida*), skin barrier, axis IL-23/Th17 and psoriasis, skin and pharmacogenetic infections of isotretinoin and antifungal

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Seborrheic dermatitis has a low genetic predisposition: the GWAS loci load is in percentile 6.3 and the "other dermatitis" score in the 15.6. Sérgio F does not have the risk alleles of **IRF4**, **IL12B** (seborrheic dermatitis locus), **MC1R** or **IL2RA**. Psoriasis is in percentile 2.4 (without the marker of HLA-C*06:02), which makes sebopsoriasis unlikely. The innate antifungal immunity is intact in the variants read: Dectin-1 without Y238X, TLR2 without R753Q and MBL2 normal. Rosacea (11.7) and atopic dermatitis (17.1) have low loci loads, and CYP2C8 is normal (isotretinoin).

Unfavourable. The score of eczema and dermatitis (any) is in percentile 90.4: a general tendency for skin inflammation and a more reactive barrier. It has **CARD14** and **KLK6/KLK7** in risk homozygosity, 1 allele in **SPINK5** (locus GWAS and E420K) and in TAP2, and is heterozygous in **TLR2 P631H** and **TLR1**. CYP2C9 is *1/*3.

What does it mean? With a low specific predisposition, Sérgio F seborrheic dermatitis is mainly explained by non-genetic factors: sebum, Malassezia, stress, sleep and drugs (rapamycin can cause seborrheic eruptions and tadalafil facial flush). The genes point to three things: caring for the barrier during isotretinoin, treating Malassezia topically and never using oral azoles with rapamycin.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality DR2 ≥ 0.8 . Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 30 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 23 of the chip, 6 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 5 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 5 PGS Catalog scores calculated locally with 1.759.100 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population (n = 633) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population (n = 633) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 1188 loci, 1106 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all scores, a higher percentile = more risk. There are no published polygenic scores specific to seborrheic dermatitis; the closest is "other dermatitis", and the loci load of the 5 section uses the specific GWAS.

Dermatitis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Eczema and dermatitis (any)	PGS000944 (Tanigawa 2022, 598 variants)	515 597 (weighted coverage 89%)	90,4	MAIS PROVABLE	unfavourable
Other dermatitis (time to the event)	PGS000927 (Tanigawa 2022, 242 variants)	212 240 (weighted coverage 88%)	15,6	MENOS PROVABLE	favourable
Atopic dermatitis	PGS002755 (Mars 2022, PRS-CS, 1.09 million variants)	1.059.974 1.090.702 (weighted coverage 96%)	64,8	TIPIC	typical

Differential diagnosis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Psoriasis	PGS002083 (Privé 2022, LDpred2, 71.7 thousand variants)	70.315 71.744 (weighted coverage 99%)	2,4	MENOS PROVABLE	favourable

Skin infections

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Surface cellulitis and abscess	PGS002081 (Privé 2022, LDpred2, 641 thousand variants)	628.084 640.921 (weighted coverage 97%)	48,8	TIPIC	typical

4. Key variants by topic

Variants with described effect. The judgment refers to seborrheic dermatitis and inflammatory skin diseases; in pharmacogenetics it refers to exposure to drugs.

Loci GWAS of seborrheic dermatitis

Loci of the largest GWAS of seborrheic dermatitis (PheCode 690.1). Many are shared with psoriasis (IL12B, CARD14, TYK2) and with pigmentation (IRF4, MC1R): the light skin and the axis IL-23/Th17 increase the probability of the disease.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
IRF4	rs12203592		C/C	chip	T (0 copy/s)	favourable	IRF4: the strongest locus of seborrheic dermatitis (also linked to pigmentation and immune response) [GWAS Catalog: Seborrheic dermatitis (PheCode 690.1), allele C, beta 0.206 [0.18-0.23] unit decrease, p=4E-71, PMID 39024449] (the reported allele C reduces the trait; here the allele of effect is the opposite, T)	Moderate
IL12B	rs12188300		A/A	chip	T (0 copy/s)	favourable	IL12B (p40 subunit of IL-12 and IL-23): shared with psoriasis [GWAS Catalog: Seborrheic dermatitis (PheCode 690.1), allele A, beta 0.215 [0.19-0.24] unit decrease, p=3E-57, PMID 39024449] (the allele A reported reduces the trait; here the allele of effect is the opposite, T)	Moderate
TAP2 (HLA)	rs241454		G/A	imputation DR2=1.00	G (1 copy/s)	unfavourable (light)	HLA region of class II (presentation of antigens) [GWAS Catalog: Seborrheic dermatitis (PheCode 690.1), allele A, beta 0.138 [0.12-0.15] unit decrease, p=2E-50, PMID 39024449] (the allele A reported reduces the trait; here the allele of effect is the opposite, G)	Moderate
MC1R	rs1805007	R151C	C/C	chip	T (0 copy/s)	favourable	MC1R R151C (light skin and freckles): shared with the rosacea [GWAS Catalog: Seborrheic dermatitis (PheCode 690.1), allele C, beta 0.172 [0.14-0.2] unit decrease, p=7E-34, PMID 39024449] (the reported allele C reduces the trait; here the allele of effect is the opposite, T)	Moderate
CARD14	rs11150849		G/G	imputation DR2=0.98	G (2 copy/s)	unfavourable	CARD14 (NF-κB signaling of the keratinocyte): shared with psoriasis [GWAS Catalog: Seborrheic dermatitis (PheCode 690.1), allele G, beta 0.089 [0.072-0.106] unit increase, p=2E-27, PMID 39024449]	Moderate
KLK6/ KLK7	rs268890		G/G	imputation DR2=0.98	G (2 copy/s)	unfavourable	Calycreins of the corneal layer [GWAS Catalog: Seborrheic dermatitis (PheCode 690.1), allele G, beta 0.111 [0.086-0.137] unit increase, p=1E-21, PMID 39024449]	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SPINK5	rs7700488		G/A	imputation DR2=0.98	G (1 copy/s)	unfavourable (light)	SPINK5 (LEKTI) [GWAS Catalog: Seborrheic dermatitis (PheCode 690.1), allele G, beta 0.075 [0.058-0.093] unit increase, p=4E-19, PMID 39024449]	Moderate
IL2RA	rs12722496		A/A	imputation DR2=1.00	G (0 copy/s)	favourable	IL-2 receiver (t-regulator cells) [GWAS Catalog: Seborrheic dermatitis (PheCode 690.1), allele A, beta 0.112 [0.086-0.137] unit decrease, p=7E-19, PMID 39024449] (the allele A reported reduces the trait; here the allele of effect is the opposite, G)	Moderate
ZMIZ1	rs1250563		C/C	chip	G (0 copy/s)	favourable	ZMIZ1 (immune transcription factor) [GWAS Catalog: Seborrheic dermatitis (PheCode 690.1), allele G, beta 0.079 [0.06-0.097] unit increase, p=2E-18, PMID 39024449]	Weak
TYK2	rs34536443	P1104A	G/G	chip	G (2 copy/s)	unfavourable	TYK2 P1104A: the function loss variant protects from psoriasis; in seborrheic dermatitis the orientation follows GWAS [GWAS Catalog: Seborrheic dermatitis (PheCode 690.1), allele G, beta 0.172 [0.13-0.21] unit increase, p=6E-17, PMID 39024449]	Moderate

Innate immunity and response to fungi (Malassezia, Candida)

Malassezia is a lipophilic yeast that lives in tallow. Seborrheic dermatitis appears when the inflammatory response to it is excessive. Dectin-1, CARD9, TLR and mannose binding lectin recognize yeasts and other skin microbes.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CLEC7A (Dectin-1)	rs16910526	Y238X	A/A	chip	C (0 copy/s)	favourable	Y238X: loss of Dectin-1 receptor, which recognizes fungal beta-glucans (Candida, Malassezia)	Moderate
CARD9	rs4077515	S12N	T/T	chip	T (2 copy/s)	without judgment	S12N: the N12 allele is associated with more inflammatory bowel disease and different antifungal responses; effect on uncertain skin	Weak
TLR2	rs5743708	R753Q	G/G	chip	A (0 copy/s)	favourable	R753Q: less TLR2 signaling (recognizable fungi and Gram+ bacteria); associated with Staphylococcus and Candida infections	Weak
TLR2	rs5743704	P631H	C/A	chip	A (1 copy/s)	unfavourable (light)	P631H: lower TLR2 signaling	Weak
TLR1	rs5743618	I602S	C/A	chip	C (1 copy/s)	unfavourable (light)	TLR1 I602S: rosacea locus; alters the response to skin bacteria [GWAS Catalog: Rosacea (PheCode 695.3), allele C, beta 0.116 [0.086-0.147] unit increase, p=6E-17, PMID 39024449]	Weak
MBL2	rs1800450	G54D	C/C	chip	T (0 copy/s)	favourable	G54D (B allele): less lectin binding to mannose, which binds to yeasts	Weak

Skin barrier

Filagrin and LEKTI (SPINK5) keep the corneal layer intact. A more fragile barrier leads to irritants and fatty acids released by *Malassezia*, and adds to the dryness caused by isotretinoin.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FLG	rs61816761	R501X	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	The so-called copy/s	without data	R501X: filagrin null mutation (the main barrier protein); ~2-4% of carriers	Strong
SPINK5	rs2303067	E420K	A/G	chip	A (1 copy/s)	unfavourable (light)	E420K: less LEKTI, more desquamation; associated with atopy	Weak
IL13	rs20541	R130Q	G/G	chip	A (0 copy/s)	favourable	R130Q: more Th2 signaling and more atopy	Moderate

IL-23/Th17 axis and psoriasis (sebopsoriasis)

Sebopsoriasis is between seborrheic dermatitis and psoriasis. The markers of HLA-C*06:02 and IL-23 axis help to understand if the psoriatic side weighs.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HLA-C*06:02 (tag)	rs10484554		C/C	chip	T (0 copy/s)	favourable	HLA-C*06:02 marker, the main genetic factor of psoriasis (OR ~2-3 by allele)	Strong
IL23R	rs11209026	R381Q	G/G	chip	A (0 copy/s)	neutral	R381Q: protective variant of IL-23 axis (less psoriasis and Crohn's disease)	Strong
IL12B	rs3212227		T/G	chip	T (1 copy/s)	unfavourable (light)	IL12B 3UTR (allele A of the gene = T in strand +, the common): psoriasis risk allele (OR ~1.4)	Moderate
TNIP1	rs17728338		G/G	chip	A (0 copy/s)	favourable	TNIP1 (NF-κB cross) [GWAS Catalog: Psoriasis, allele A, OR 1.59, p=1E-20, PMID 19169254]	Moderate
ERAP1	rs30187	K528R	C/C	chip	T (0 copy/s)	favourable	ERAP1 K528R (interacts with HLA-C*06:02)	Moderate

Pharmacogenetics: isotretinoin, antifungals and rapamycin

Isotretinoin is mainly metabolized by CYP2C8, and also by CYP2C9 and CYP3A4. Oral azoles (ketoconazole, itraconazole, fluconazole and others) inhibit CYP3A and greatly increase the levels of rapamycin and tadalafil.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CYP2C8	rs11572080	*3 (R139K)	C/C	imputation DR2=1.00	T (0 copy/s)	favourable	CYP2C8*3 (R139K): isotretinoin and tretinoin are mainly metabolized by CYP2C8	Moderate
CYP2C8	rs10509681	*3 (K399R)	T/T	chip	C (0 copy/s)	favourable	CYP2C8*3 (K399R)	Moderate
CYP2C8	rs1058930	*4	G/G	chip	C (0 copy/s)	favourable	CYP2C8*4 (I264M)	Moderate
CYP2C9	rs1799853	*2	C/C	chip	T (0 copy/s)	favourable	CYP2C9*2: metabolism of fluconazole and NSAIDs	Strong
CYP2C9	rs1057910	*3	A/C	chip	C (1 copy/s)	unfavourable (light)	CYP2C9*3: very reduced function	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CYP3A5	rs776746	*3	C/T	chip	C (1 copy/s)	without judgment	CYP3A53: with 1/*3 there is partial expression of CYP3A5 (rapamycin; oral azoles inhibit CYP3A)	Strong

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each disease, with each allele oriented towards the risk, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Seborrheic dermatitis	26 (25)	15 de 50	25	6,3	favourable
Rosacea	9 (8)	5 de 16	8	11,7	favourable
Psoriasis	746 (728)	746 de 1456	728	24,2	typical
Atopic dermatitis	407 (345)	274 de 690	345	17,1	favourable

6. Favourable

- **Seborrheic dermatitis with low predisposition:** loci load in percentile 6.3; IRF4 C/C, IL12B rs12188300 A/A, MC1R R151C C/C, IL2RA A/A and ZMIZ1 C/C without the risk allele "other dermatitis" in percentile 15.6. **Moderate.**
- **Psoriasis very unlikely:** score in percentile 2.4; marker of HLA-C*06:02 absent (C/C); TNIP1 and ERAP1 without risk; loci load in 24.2. **Strong.**
- **Innate intact antifungal immunity** in the variants read: Dectin-1 Y238X absent (A/A), TLR2 R753Q absent, MBL2 G54D absent. **Moderate.**
- **Rosacea and atopic dermatitis:** loci loads in the 11.7 and 17.1 percentiles; IL13 R130Q absent. The genomic score of atopic dermatitis is typical (64.8). **Moderate.**
- **Bacterial infections of the skin** (cellulitis and abscess): percentile 48.8, typical. **Weak.**
- **Normal CYP2C8** (without *3 or *4): typical metabolism of isotretinoin. **Moderate.**

7. Unfavourable

- **Eczema and dermatitis (any)** in percentile 90.4: more reactive skin and prone to contact dermatitis and eczematous, which isotretinoin dryness may reveal. **Moderate.**
- **Loci of the barrier and desquamation:** CARD14 G/G and KLK6/KLK7 G/G in risk homozygosity; SPINK5 rs7700488 G/A and E420K A/G with 1 allele. **Moderate / weak.**
- **TYK2 G/G:** without the protective allele P1104A; **IL12B rs3212227 T/G** with 1 psoriasis risk allele (the psoriasis score remains very low). **Weak.**
- **TLR2 P631H C/A** and **TLR1 C/A** heterozygotes: a slightly different response to skin bacteria and yeasts. **Weak.**
- **CYP2C9 *1/*3:** fluconazole and NSAIDs spend more time in the body. Strong (pharmacogenetic).
- **No judgment:** CARD9 S12N T/T (N12/N12); CYP3A5 *1/*3. FLG R501X is not read reliably (insufficient imputation).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Dysbiosis	Skin microbiome and Malassezia: intact antifungal innate immunity (Dectin-1, MBL2); TLR2 P631H and TLR1 heterozygotes	Neutral
Chronic Inflammation	IL-23/Th17 axis: psoriasis in percentile 2.4, HLA-C*06:02 absent; seborrheic dermatitis with loci in 6.3	Favourable
Inflammaging	Eczema and dermatitis (any) in percentile 90.4	Unfavourable (mild)
Immunosenescence	Yeast recognition (Dectin-1, CARD9, MBL2) without function loss variants	Favourable
Remodelling of the extracellular matrix	Barriers and desquamation: CARD14 and KLK6/KLK7 in risk homozygosity; SPINK5 with 1 allele	Unfavourable (mild)
Nutrient Sensing Deregulation	Sebocyte and mTORC1: without genetic marker; rapamycin and isotretinoin act in this pathway	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Topical antifungal of maintenance** (with dermatologist): ketoconazole 2% or cyclopyrox in shampoo or cream 2–3×/week in outbreaks and 1×/week in maintenance; zinc pyrithione or selenium sulphide as an alternative. With low genetic predisposition, controlling *Malassezia* and sebum is the one that most changes the course of the disease. *Suggestion (not applied)*.
2. **Never oral azoles with rapamycin:** ketoconazole, itraconazole, voriconazole and posaconazole (and, to a lesser extent, fluconazole) inhibit CYP3A and greatly increase the levels of rapamycin and tadalafil. Fluconazole, with CYP2C9 *1/*3, also stays longer in the body. If one day an oral antifungal is needed, decide with the doctor and adjust or suspend rapamycin. *Suggestion (not applied)*.
3. **Barriers during isotretinoin:** The eczema score in the ~the 90th percentile and the loci CARD14, KLK and SPINK5 indicate a more reactive barrier. Use emollient with ceramides, smooth cleaning without sulfates and warm water. On the face, prefer moisturizers with squalane or mineral oil, which do not feed *Malassezia*. *Suggestion (not applied)*.
4. **Weekly isotretinoin:** The normal CYP2C8 does not require dose adjustment because of the genes. The lowest seborrheic dermatitis helps, but dryness can bring eczematous dermatitis. Distinguish both with the dermatologist, because the treatment is different. *Suggestion (not applied)*.
5. **In outbreaks:** short course of low-power corticosteroids, or a topical calcineurin inhibitor (tacrolimus, pimecrolimus), or roflumilast in foam. On the face, avoid potent corticosteroids and continuous use. *Suggestion (not applied)*.
6. **Register the triggers:** 1–2 outbreaks after each rapamycin dose, flushing with tadalafil, stress and short sleep. *Suggestion (not applied)*.
7. **Nutrition, as it is already in the plan:** Total zinc of 20–25 mg/day with copper (the zinc deficit gives seborrheic type eruptions), B6 ≤ 5 mg (high doses give acneiform eruptions), omega-3 to 3.2–4 g/day, high fiber (the better) and low free sugars. *Suggestion (not applied)*.
8. **Daily photoprotection:** isotretinoin increases photosensitivity. If it is considered UV phototherapy for dermatitis, do so with the dermatologist and with the examination of the skin in day. *Suggestion (not applied)*.

Not recommended: oral azoles with rapamycin; oils rich in oleic acid in the skin (azeite, coconut), which feed *Malassezia*; strong topical retinoids over isotretinoin; continuous potent corticosteroids on the face.

10. Limitations

- Seborrheic dermatitis has few genetic studies. There is only one large GWAS (clinical record phenotype) and there are no published specific polygenic scores.
- The chip reads only a few variants of the filagrine. R501X does not have reliable imputation and the other null mutations (e.g., 2282del4) do not read.
- The microbiome of the skin depends mainly on the environment, the tallow and the applied products. The host genes explain little of the microbial composition.
- The loci load sum of loci in linkage imbalance and shared with pigmentation. It is a descriptive complement and does not replace the polygenic scores.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromics, strand + (GWAS Catalog standard and the harmonized files of PGS Catalog) is assumed. In the loci of GWAS Catalog with protective effect, the allele of effect is exchanged by the risk allele.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS000944 and PGS000927 (Tanigawa 2022), PGS002755 (Mars 2022), PGS002083 and PGS002081 (Privé 2022). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: GWAS of clinical phenotypes of 2024 (seborrheic and rosacea dermatitis), Ferwerda 2009 (Dectin-1), Palmer 2006 (filagrin), Tsoi 2012 and 2017 (psoriasis), reviews of seborrheic dermatitis and Malassezia, and the labels of isotretinoin and sirolimus. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Loci GWAS of seborrheic dermatitis	IRF4	rs12203592	C/C	chip	T	0	strand +	without the risk allele
Loci GWAS of seborrheic dermatitis	IL12B	rs12188300	A/A	chip	T	0	strand + (palindromic; assumed +)	without the risk allele
Loci GWAS of seborrheic dermatitis	TAP2 (HLA)	rs241454	G/A	imputation DR2=1.00	G	1	strand +	1 copy of risk allele
Loci GWAS of seborrheic dermatitis	MC1R	rs1805007	C/C	chip	T	0	strand +	without the risk allele
Loci GWAS of seborrheic dermatitis	CARD14	rs11150849	G/G	imputation DR2=0.98	G	2	strand + (palindromic; assumed +)	2 risk allele copies
Loci GWAS of seborrheic dermatitis	KLK6/KLK7	rs268890	G/G	imputation DR2=0.98	G	2	strand +	2 risk allele copies
Loci GWAS of seborrheic dermatitis	SPINK5	rs7700488	G/A	imputation DR2=0.98	G	1	strand +	1 copy of risk allele
Loci GWAS of seborrheic dermatitis	IL2RA	rs12722496	A/A	imputation DR2=1.00	G	0	strand +	without the risk allele
Loci GWAS of seborrheic dermatitis	ZMIZ1	rs1250563	C/C	chip	G	0	strand + (palindromic; assumed +)	without the risk allele
Loci GWAS of seborrheic dermatitis	TYK2	rs34536443	G/G	chip	G	2	strand + (palindromic; assumed +)	2 risk allele copies
Innate immunity and response to fungi (Malassezia, Candida)	CLEC7A (Dectin-1)	rs16910526	A/A	chip	C	0	strand +	without the risk allele
Innate immunity and response to fungi (Malassezia, Candida)	CARD9	rs4077515	T/T	chip	T	2	strand +	2 copy of the effect allele
Innate immunity and response to fungi (Malassezia, Candida)	TLR2	rs5743708	G/G	chip	A	0	strand +	without the risk allele
Innate immunity and response to fungi (Malassezia, Candida)	TLR2	rs5743704	C/A	chip	A	1	strand +	1 copy of risk allele
Innate immunity and response to fungi (Malassezia, Candida)	TLR1	rs5743618	C/A	chip	C	1	strand +	1 copy of risk allele
Innate immunity and response to fungi (Malassezia, Candida)	MBL2	rs1800450	C/C	chip	T	0	strand +	without the risk allele
Skin barrier	FLG	rs61816761	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	A			SNP not found in the data
Skin barrier	SPINK5	rs2303067	A/G	chip	A	1	strand +	1 copy of risk allele
Skin barrier	IL13	rs20541	G/G	chip	A	0	strand +	without the risk allele
IL-23/Th17 axis and psoriasis (sebopsoriasis)	HLA-C*06:02 (tag)	rs10484554	C/C	chip	T	0	strand +	without the risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
IL-23/Th17 axis and psoriasis (sebopsoriasis)	IL23R	rs11209026	G/G	chip	A	0	strand +	without the protective allele (common genotype)
IL-23/Th17 axis and psoriasis (sebopsoriasis)	IL12B	rs3212227	T/G	chip	T	1	strand +	1 copy of risk allele
IL-23/Th17 axis and psoriasis (sebopsoriasis)	TNIP1	rs17728338	G/G	chip	A	0	strand +	without the risk allele
IL-23/Th17 axis and psoriasis (sebopsoriasis)	ERAP1	rs30187	C/C	chip	T	0	strand +	without the risk allele
Pharmacogenetics: isotretinoin, antifungals and rapamycin	CYP2C8	rs11572080	C/C	imputation DR2=1.00	T	0	strand +	without the risk allele
Pharmacogenetics: isotretinoin, antifungals and rapamycin	CYP2C8	rs10509681	T/T	chip	C	0	strand +	without the risk allele
Pharmacogenetics: isotretinoin, antifungals and rapamycin	CYP2C8	rs1058930	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele
Pharmacogenetics: isotretinoin, antifungals and rapamycin	CYP2C9	rs1799853	C/C	chip	T	0	strand +	without the risk allele
Pharmacogenetics: isotretinoin, antifungals and rapamycin	CYP2C9	rs1057910	A/C	chip	C	1	strand +	1 copy of risk allele
Pharmacogenetics: isotretinoin, antifungals and rapamycin	CYP3A5	rs776746	C/T	chip	C	1	strand +	1 copy of effect allele